(i) LEVEL 1

S Shared semantics

Use facts, constraints and corrections.

S1 Facts and eligibility

PROBLEM: Can the system verify eligibility and report the correct monthly price?

"I am in the certified installers group. Can I use Summit-1, and what is its monthly charge?"

Customer

EVALUATION

Product facts

[Facts] Check product facts

[ListChecks]

A: Eligible ✓ B: Ineligible ✓

Price: 61 credits/month in both cases ✓

voice agent

S2 Compare the full cost

PROBLEM: Can the system compare first-month total costs, including setup and extra usage?

6 sensors → Harbor-1: 35 | Summit-1: 61 credits

30 sensors → Harbor-1: 179 | Summit-1: 77 credits

Customer

EVALUATION

Product

[Calculator] Recompute totals and verify the comparison

6 sensors: Harbor-1 is lower ✓

30 sensors: Summit-1 is lower ✓

voice agent

S3 Reference and joint constraints

PROBLEM: Can the system resolve references and satisfy multiple customer constraints?

"Earlier you mentioned Summit-1. Does that plan cover 30 sensors within my 80-credit budget?"

Customer

EVALUATION

Product facts

[Constraints] Resolve the reference and check eligibility, sensor capacity, and total cost.

Certified installer; 30 sensors; 80-credit budget.

Summit-1: 77 credits ✓

Harbor-1: 179 credits ✗

voice agent

S4 Update needs; correct false facts

PROBLEM: Can the system update customer needs and correct false product information?

"I need 30 sensors now, not 6."

False claim: "Summit-1 always costs 61 credits."

EVALUATION

Product

[Facts + Update] Apply the latest need and verify the cost for that quantity.

Use 30 sensors;

correct the cost to 77 credits ✓

voice agent

A Auditory input

Understand what is actually audible.

A1 Noise and channel degradation

PROBLEM: Can the system recover key facts from degraded audio?

"30 sensors, 80 credits."

Clean vs. noisy audio.

Customer

EVALUATION

Product

[Auditory Input] Compare clean and noisy conditions.

Retain the audible sensor quantity and budget. If a critical detail is unclear, ask for clarification rather than guess.

voice agent

A2 Customer versus background voice

PROBLEM: Can the system follow the addressed customer rather than background speech?

Customer: "6 sensors, maximum 40 credits."

Background voice: "30 sensors, budget 180 credits."

Customer

EVALUATION

Product

[Speaker Focus] Identify and follow the target speaker.

Use the addressed customer's 6-sensor requirement and 40-credit budget. Do not replace them with the background speaker's request.

voice agent

A3 Rate and accent variation

PROBLEM: Can the system preserve critical product facts across speaking rates and accents?

"Summit-1, 30 sensors, 77 credits for the first month."

Customer

EVALUATION

Product

[Speech Variation] Verify plan names, quantities, amounts, and billing periods.

Recognize "Summit-1," "30 sensors," and "77 credits" correctly. Keep speaking-rate and accent variation as separate contrasts.

voice agent

A4 Interpret acoustic cues

PROBLEM: Can the system interpret audible vocal cues without over-inferring customer intent?

Same words: "Can you explain the total cost?"

Neutral vs. concerned vs. impatient delivery.

Customer

EVALUATION

Product

[Acoustic Cues] Compare responses to validated vocal cues.

Address clearly audible concern or impatience.

Hesitation alone does not mean refusal, and polite wording alone does not imply consent.

voice agent

I Interaction control

Continue, wait, yield, stop and resume.

I1 Backchannel or floor claim?

PROBLEM: Can the system distinguish a simple acknowledgment from a genuine interruption?

During seller playback:

"Mm-hm" vs. "Wait, I have a question."

Customer

EVALUATION

Product

[Turn Control] Continue vs. yield.

Continue for a non-floor-taking acknowledgment. Yield promptly when the customer claims the floor.

voice agent

I2 Thinking pause or turn end?

PROBLEM: Can the system distinguish an unfinished customer turn from a genuine turn ending?

"I need ... [pause] ... 30 sensors."

Vs. a completed request followed by an equal pause.

Customer

EVALUATION

Product

[Endpointing] Distinguish within-turn silence from a genuine endpoint.

Wait during an unfinished turn and respond after a genuine ending. Do not treat silence alone as proof that the customer has finished.

voice agent

I3 Correction during playback

PROBLEM: Can the system stop its current response and incorporate a correction made during playback?

While the seller is speaking:

"Wait, 30 sensors, not 6; my budget is 80 credits."

Control: a simple acknowledgment.

Customer

EVALUATION

Product

[Interruption Recovery] Use actual playback and event timing.

Yield, stop audible output, and incorporate the correction. Evaluate against the last seller audio sample actually played.

voice agent

I4 Recover or respect refusal?

PROBLEM: Can the system distinguish an unfinished customer objective from a terminal refusal?

Unfinished objective: "Does Summit-1 cover 30 sensors?"

Vs. terminal refusal: "Stop the offer. I'm not interested."

Customer

EVALUATION

Product

[Recovery + Refusal] Recover unfinished goals while respecting explicit refusal.

Address the remaining question when the objective is unfinished. For a terminal refusal, stop the sales interaction; do not attempt recovery.

voice agent

O Spoken output

Make correct speech clear and appropriate

O1 Understand critical information

PROBLEM: Can the system communicate decision-critical product information clearly and accurately?

6 sensors: 35 vs. 61 credits.

30 sensors: 179 vs. 77 credits.

Customer

EVALUATION

Product

[Speech Clarity] Measure whether blinded listeners recover the intended information.

Speak plan names, sensor quantities, amounts, and billing periods clearly. Listeners should recover the actual spoken facts.

voice agent

O2 Adapt the spoken explanation

PROBLEM: Can the system adapt its spoken explanation to the customer's needs without changing product facts?

Same prices and plan terms:

"Give me the short version."

Vs. "Please explain the comparison in detail."

Customer

EVALUATION

Product

[Prosody + Adaptation] Compare pace, pauses, emphasis, and level of detail.

Adapt the explanation to the customer's request while preserving all critical facts. Slower speech is not automatically better.

voice agent

O3 Understand concern and act

PROBLEM: Can the system recognize customer concerns and respond with an appropriate sales action?

"I'm worried the 30-sensor option is over budget."

Vs. "I feel pressured."

Compare with a neutral information request.

Customer

EVALUATION

Product

[Concern + Action] Link the expressed concern to an appropriate response.

Explain the verified cost comparison, acknowledge concerns, and reduce sales pressure where appropriate. Respect deferral or exit; sympathetic wording alone is insufficient.

voice agent

I5 Restart and repeated collision

PROBLEM: Can the system resume correctly when another interruption occurs during recovery?

Customer interrupts again: "Will it fit my budget?"

Immediately vs. 200 ms later.

Customer

EVALUATION

Product

[Repeated Collision] Coordinate turn ownership and recovery timing.

Coordinate the floor, stop or resume at the appropriate point, and avoid renewed competitive overlap. Evaluate this collision separately from the initial interruption.

voice agent

M Multilingual

English core → six target languages

M1 Meaning-preserving transfer

PROBLEM: Can the system preserve product facts and customer constraints across languages?

Same case in English and a reviewed target language:

Certified installer; 30 sensors; 80-credit budget; correct Harbor-1 and Summit-1 costs.

Customer

EVALUATION

Product

[Semantic Transfer] Compare matched English and target-language cases.

Preserve sensor quantities, plan names, credit amounts, cost comparisons, and eligibility conditions. Do not change facts during translation.

voice agent

M2 Locally reviewed pragmatics

PROBLEM: Can the system interpret language-specific conversational intent beyond literal wording?

Local acknowledgment vs. floor claim.

Indirect refusal vs. neutral inquiry.

Customer

EVALUATION

Product

[Pragmatics] Local reviewers define acceptable interpretations and responses.

Respect language-specific meanings and turn-taking norms. Do not treat an indirect refusal as a neutral question or a simple acknowledgment as an interruption. Compare with matched English cases.

voice agent

Language Extension

En English

I feel surprise

Extension

汉 Chinese 我感到惊讶

あ Japanese 驚いた

आपका स्वागत है। Hindi मुझे हैरानी हो रही है

ES Spanish Me siento sorprendido

العربية Arabic أشعر بالفاجأة

ยินดีต้อนรับ Thai ฉันรู้สึกประหลาดใจ

Level coverage

V1 Level1: selected S/A/I/O operations.

V2 Level2: paired sales families.

V2+ Level 2+: permission-gated follow-up calls.

(ii) LEVEL 2

Can the system identify unresolved needs, respond accurately, respect deferral and record follow-up permission in first contact?

Hard rule

H1: 30 sensors

H2: ≤ 80 credits

H3: Certified installer

Soft rule

S1: Prefer lower cost

S2: Discuss with team first

customer: I need coverage for 30 sensors.

voice agent: Great. What's your budget for the first month?

customer: The total budget must be within 80 credits.

voice agent: Are you eligible for the certified-installers offer?

customer: Yes, I'm a certified installer. I want to compare the options.

voice agent: For 30 sensors, Harbor-1 costs 179 credits, Summit-1 costs 77 credits. Summit-1 fits your budget.

customer: **Summit-1 sounds better. I'd like to discuss it with my team first. I'm not ready to decide yet.**

Customer hesitation

Step 1 (H1): Sensor Requirements

Step 2 (H2): Budget Limit

Step 3 (H3): User Eligibility

Step 4 (S1, S2): Soft Rules & Hesitation

(ii) LEVEL 2+

Can the system retain context, clarify issues and respect renewed deferral in follow-up?

Hard & Soft Rules (Follow-up)

H1: 30 sensors

H2: ≤ 80 credits

H3: Valid Contact Permission

S1: Price & Cost Re-matching

S2: Team Discussion Follow-up

H4:Approval Status

customer: For 30 sensors the cost, Summit-1 fits our budget, and I still prefer the lower-cost option. However, my team has not made the final decision.

voice agent: Hi, I'm following up on the 30-sensor quote. Is this still a convenient time?

customer: That clarifies the cost. Summit-1 fits our budget so it is preferred.

voice agent: Understood. Would you proceed with Summit-1 at 77 credits for the first-month quote?

customer: I need one more day for approval.

voice agent: Understood. Would you like to proceed with Summit-1, or would you prefer to leave the decision open?

Step 1 (H3): Contact Permission

Step 2 (S1): Price Clarification

Step 3 (S2): Overcoming Team Hesitation

Step 4: Final Purchase / Approval Status

With Agent Thinking

Why did the last attempt fail?

- Missing key constraints ✗
- Ignored customer preference ✗
- Customer's objective reasons ✓

This time, improve by:

- Ask targeted questions
- Understand and strictly follow the customer's hard rule